

Which Automatic Image Registrations Can a Microscopist Trust? A Label-Free Test on a Correlative Microscopy Benchmark

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Abstract—Correlative microscopy combines images of one specimen taken with different instruments, which first requires registration: finding the transformation that maps one image onto the other. Foundation image matchers can register images automatically, but on the AmalgaMatch benchmark of 187 materials-science image pairs they succeed on only about a quarter, and they give no warning when they fail. I tested whether scores that need no ground truth can tell which registrations to trust. Each pair was registered 15 ways, and three scores were compared: the retained fraction (the share of proposed point matches that survive robust fitting), the agreement between registrations, and a combination of the two. For MatchAnything-RoMa, the retained fraction separated successes from failures with an area under the receiver operating characteristic curve of 0.85 (0.93 at the stricter 8-pixel criterion of earlier work), and accepting only the highest-scoring quarter of pairs raised the share of correct registrations from 24 % to 57 %. A cut-off chosen on two task groups kept every correct registration in the other four, but the share of accepted registrations that were correct fell from 71 % to 31 %, and all 42 wrongly accepted pairs were transmission electron microscope images of one alloy. Combining the retained fraction with agreement between registrations did not improve prediction, and choosing one registration per pair did not beat the best single method. The hypotheses and tests were fixed and published before the runs reported here.

INTRODUCTION

Correlative microscopy places images of the same region of a specimen, taken with different instruments, on one coordinate grid, so that every position carries several kinds of measurement, for example crystal orientation from electron backscatter diffraction (EBSD) and surface contrast from a scanning electron microscope (SEM) [1]. The first step is registration: finding the transformation that maps the image with the smaller field of view (here called the narrow image) into the one with the larger field of view (the wide image). Registration by hand is slow. Image matchers pretrained on large collections of photographs, such as RoMa [3] and its version retrained on images from different sensors, MatchAnything-RoMa [4], can do it automatically. On AmalgaMatch, a public benchmark of 187 correlative image pairs from materials science [1], [2], they register only about one pair in four to within 20 pixels [5].

These matchers also do not report failure. RoMa and MatchAnything-RoMa, as configured, return 10,000 point matches (correspondences: pairs of points believed to show the same feature) for every pair, whether or not the two images show the same place, and robust fitting always finds some that agree [5]. A researcher who runs automatic registration on a new batch of images therefore cannot tell which results to keep.

Durmaz et al. [2] found that the share of correspondences surviving robust fitting separated good from bad registrations on 48 pairs from four subsets of the benchmark, and proposed

using it to choose between image pairs of different modalities and to send doubtful cases to a human. This paper tests that idea on the whole benchmark. **H1**: the retained fraction predicts success on all pairs, and a cut-off chosen on the kinds of pairs Durmaz et al. studied still works on the other kinds. **H2**: combining the retained fraction with the agreement between several registrations of the same pair predicts success better than the retained fraction alone. **H3**: extending their proposal from modality pairs to registration methods, choosing one registration per pair with these scores registers more pairs than the best single method. The hypotheses, statistics and pass criteria were committed to a public repository before the runs reported here [6]. A pilot on earlier runs of the same pairs, in which the number of surviving correspondences gave an area under the curve (defined below) of 0.83 for MatchAnything-RoMa and 0.94 for RoMa, motivated the design and the pass bar for H1.

METHODS

Data. AmalgaMatch [1] contains 187 image pairs from 35 scenes, 19 combinations of material and instruments, and six task groups (for example serial sectioning, fracture surfaces, and dislocation imaging in the transmission electron microscope, TEM). Every pair has hand-annotated corresponding points, used only to score results, never to produce them. The data are released under a CC BY 4.0 licence.

Candidate registrations. Each pair was registered 15 ways (the voting pool). Five matchers were applied to the whole images: the classical SIFT [7], LoFTR [8], RoMa [3], and the MatchAnything versions of RoMa and of Efficient LoFTR [4]. RoMa and MatchAnything-RoMa were also run inside a coarse-to-fine search that zooms into the wide image [5], and on four label-free transformations of the images: the narrow image with inverted contrast, the narrow image with its brightness histogram matched to the wide image, both images after local contrast equalization (CLAHE), and both images replaced by their gradient magnitude. As a control for random variation, RoMa and MatchAnything-RoMa were rerun with five further random seeds; these ten runs never vote. Each set of correspondences was fitted with the robust estimator MAGSAC++ [9] (5.5-pixel threshold), which repeatedly fits a transformation to small random subsets of correspondences and keeps the one that the most correspondences agree with (its inliers). The transformation is affine (shift, rotation, scale and shear) or a homography (which also allows perspective), following the benchmark protocol [1].

Label-free scores. Three scores were computed without any annotated points. **S1**, the retained fraction [2], is the number of inliers divided by the number of correspondences proposed. **S2** measures agreement. For a candidate and each other member of the voting pool, both transformations are applied to a 5×5 grid of points spanning the narrow image, and the mean distance between the two mapped grids is taken. **S2** is minus the median of these distances, so a candidate that lands where most others land scores high. **S3** averages the percentile ranks of **S1** and **S2**, ranked over all pair and candidate combinations in the voting pool.

Evaluation. A registration's error is the mean distance, in pixels of the wide image, between where it maps the annotated points and where they truly lie, with no refinement step. A registration succeeds if this error is at most 20 pixels.

The best a single global transformation can do is estimated by fitting a homography to each pair's own annotated points. How well a score separates successes from failures is measured by the area under the receiver operating characteristic curve (AUROC): the probability that a randomly chosen success scores higher than a randomly chosen failure, where 0.5 is chance and 1 is perfect. Because pairs from the same scene share images, 95 % confidence intervals come from resampling whole scenes with replacement 10,000 times. Paired differences in success counts are tested with the exact McNemar test [10], which uses only the pairs on which two methods disagree. For the transfer test, a cut-off on S1 was chosen by Youden's index [11] (the cut-off that maximizes the share of successes accepted minus the share of failures accepted) on the two task groups Durmaz et al. studied, same-slice and serial sectioning, and applied unchanged to the other four. For triage, pairs are ranked by score and the highest-scoring ones accepted. Results outside the pre-registered list are labelled exploratory.

RESULTS

The retained fraction flags failures (H1). MatchAnything-RoMa registered 44 of the 187 pairs to within 20 pixels. A homography fitted to the annotated points reached 156, an optimistic ceiling because it is scored on the points it was fitted to. The retained fraction separated MatchAnything-RoMa's successes from its failures with an AUROC of 0.85 (95 % confidence interval 0.72 to 0.98), meeting the pre-registered bar of 0.80. At 10 pixels the AUROC was 0.90 (0.80 to 0.97), and at the 8-pixel criterion of Durmaz et al., who reported 0.905 on 48 pairs [2], it was 0.93 (0.84 to 0.98). For RoMa, a secondary comparison, it was 0.94. Within the 4 task groups that had both successes and failures, the AUROC ranged from 0.81 to 1.00 (exploratory; the highest from 13 pairs). None of the 31 slip-partitioning pairs was registered, and all 6 fracture-surface pairs were (Fig. 2).

The cut-off chosen on the 68 same-slice and serial-sectioning pairs, a retained fraction of 0.171, was applied unchanged to the 119 pairs of the other four groups. It accepted 61 of them and kept every correct registration. Of the accepted registrations, 31 % were correct, against 16 % of all registrations in these groups (difference 15 percentage points, confidence interval 4 to 76), which passes the pre-registered test.

The pass is weak. The share correct among accepted registrations fell from 71 % in the groups used to choose the cut-off to 31 %; the interval is wide because the 119 pairs come from only 13 scenes; and the cut-off accepted 42 % of the failures. All 42 wrongly accepted pairs were TEM images of dislocations in one alloy, MoTaTiZrHf, from 3 scenes (exploratory); 66 of the benchmark's 69 TEM pairs come from this alloy. On these pairs the fit kept a large share of the correspondences but missed by 21 to 527 pixels (median 187; Fig. 1c, Fig. 2): the matches agreed with one another but not with the truth. The cause was not tested, and because only one alloy is involved, it may lie in the specimen rather than the microscope.

Triage. Ranking all pairs by the retained fraction and accepting the top quarter raised the share of correct registrations from 24 % to 57 % (confidence interval 33 % to 97 %); accepting the top half or top three quarters gave 41 % and 31 % (Fig. 3a). The area under the risk-coverage curve,

the average failure rate among accepted registrations over all acceptance levels, was 0.54 (0.29 to 0.71), against 0.76 for a random order. In exploratory counts, the 69 lowest-scoring registrations were all failures, and every correct registration ranked in the top 63 %. At the other end, 20 of the 47 registrations in the top quarter were failures, all from the same TEM alloy.

Agreement did not improve prediction (H2). For MatchAnything-RoMa, agreement alone gave an AUROC of 0.72, lower than the retained fraction by -0.14 (-0.28 to -0.03), and the combined score gave 0.80, a difference of -0.06 (-0.13 to -0.01). Pooled over all 15 candidates, the combined score differed from the retained fraction by -0.01 (-0.06 to $+0.02$). H2 required the combined score to do better. It did not in either comparison, and for MatchAnything-RoMa it did significantly worse. One reason is that the candidates were not independent: 12 of the 15 are variants of RoMa, and they tend to fail together. On 107 pairs no candidate succeeded, and of the 62 pairs on which at least half of the candidates agreed to within 20 pixels, 31 had no correct candidate at all (exploratory).

Choosing per pair did not beat the best single method (H3). The best single method, MatchAnything-RoMa inside the zoom search, registered 47 pairs. Choosing per pair by the retained fraction registered 42 (6 pairs gained, 11 lost, McNemar $p = 0.332$; difference -6 to 1 percentage points), by agreement 40, and by the combined score 47 (Fig. 3b). H3 is not supported. An oracle that always picked a correct registration when one existed would have registered 80. As a control, picking the best of six runs of one matcher by the retained fraction gained 2 pairs for MatchAnything-RoMa and 4 for RoMa.

Exploratory follow-ups suggest two reasons. First, the retained fraction is not comparable across matchers. SIFT proposed far fewer correspondences (median 99, against 10,000 for the RoMa family), and a fit to few points keeps a larger share of them (median retained fraction 0.31 against 0.13). Choosing by the retained fraction picked SIFT on 79 pairs, and 9 of the 11 pairs it lost were SIFT picks. Within a single pair, the fraction usually did prefer a correct registration over a failed one (median within-pair AUROC 0.86 over the 77 pairs that had both), and choosing only among the 12 RoMa-family candidates registered 56 pairs (14 gained, 5 lost, $p = 0.06$); that rule needs its own pre-registered test. Second, much of the oracle's lead rests on unstable successes. Of the 33 pairs that only the oracle registered, 16 were registered by at most two of the 15 candidates. Across its six runs, MatchAnything-RoMa registered 51 different pairs at least once, and 16 of those succeeded in some runs and failed in others.

DISCUSSION

Any registration pipeline that uses robust fitting already computes the retained fraction, so using it needs no extra computation. On the full benchmark it separated good from bad registrations as well as Durmaz et al. found on four subsets [2]. Ranking by it was most reliable at the bottom of the list: here the lowest-scoring registrations could have been discarded or rerun without losing a single correct one. The top of the list was less safe. A cut-off learned on one kind of image pair accepted many confident failures on another kind, the TEM images of one alloy, so the fraction should be

calibrated for each kind of image, or its high scores checked by hand on a new kind, before it is trusted. It should not be compared across matchers that propose very different numbers of correspondences, and agreement among variants of one matcher is not a check on correctness.

The study uses one benchmark; its 35 scenes make the confidence intervals wide, and the transfer test rests on 13 scenes. Each candidate was run once, and MatchAnything-RoMa's six runs registered between 41 and 44 pairs, so the best single method, chosen on the same data, led the next candidates by less than this run-to-run spread. The 20-pixel criterion was applied to wide images from 293 to 10,147 pixels across, so it is stricter for larger images; results at 8 and 10 pixels are given. The image transformations were chosen for simplicity, not tuned.

For materials scientists using automatic registration, these results support a simple procedure: record the retained fraction, sort a batch by it, rerun or discard the lowest-scoring registrations, and check the high-scoring ones by hand on any kind of image for which the score has not yet been validated.

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